UiPath The Foundation of Innovation™

Introduction to Automation

Instructor-led training: Student guide

Tap into the potential of automation: Learn the basics today!



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Hello!

Welcome! If you're new to automation and exploring the UiPath Automation Cloud platform or just looking for career opportunities in this field, you're in the right place! In fact, this course is for all of you!

Fasten your seatbelts and get ready to go on a ride into the core aspects of automation using UiPath, learn about the practical benefits for you and your organization, and understand different human-robot collaboration scenarios. Don't miss the chance to create your very first automation project as you take your first steps into the world of automation. Let's get started!

What you'll be empowered with at the end of this course…

At the end of this course, you should be able to:

- Describe automation fundamentals, including UiPath automation models, and effectively share the benefits with colleagues across departments.
- Describe different collaborative scenarios between humans and robots in an automated enterprise, as well as the roles that support the automation journey and the roles you can take in this context.
- Create a simple automation project in Studio Web using a template.
- Assess the applicability of automation for personal tasks and identify potential tasks/processes from daily work that can be automated.

Logistics

To get started, download the **Resources Repository** zip folder and keep it handy. Your facilitator will provide you with instructions on how and when to use the files in this folder.



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What is automation?

Once upon a time, people used to wash clothes by hand. This was a time-consuming and physically demanding task. With the invention of the washing machine, you can easily and quickly clean your clothes with just the push of a button. Instead of spending hours washing clothes by hand, you can now spend that time doing something else you enjoy, like spending time with family, reading a book, or pursuing a hobby.

The washing machine is a great example of how automation can make our lives easier and more efficient. With automation, we can automate repetitive tasks and free up time and energy for more important things. Similarly, software automation takes care of the repetitive and tedious tasks that you perform on your computer so that it frees up time for you to focus on more important work.

In short, automation eliminates repetitive tasks, freeing us to focus on essential work and lead more productive lives. Let's discover how!

How do you define automation?

Automation, in its essence, involves leveraging technology, software, or machinery to carry out tasks that would typically require human effort. The beauty of automation lies in its ability to minimize or eliminate the need for constant human intervention, allowing processes to run smoothly and efficiently.



Automation is a game changer because it revolutionizes the way businesses and individuals operate. It leads to increased efficiency and improved accuracy by reducing human errors and streamlining processes. Automation allows organizations to cut costs, enhance scalability, and provide 24x7 operations. By automating repetitive tasks, employees can focus on more strategic and creative responsibilities, leading to higher job satisfaction.

It's like having a reliable digital assistant that takes care of the mundane, allowing you to channel your energy and expertise where it matters most.

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The story of work

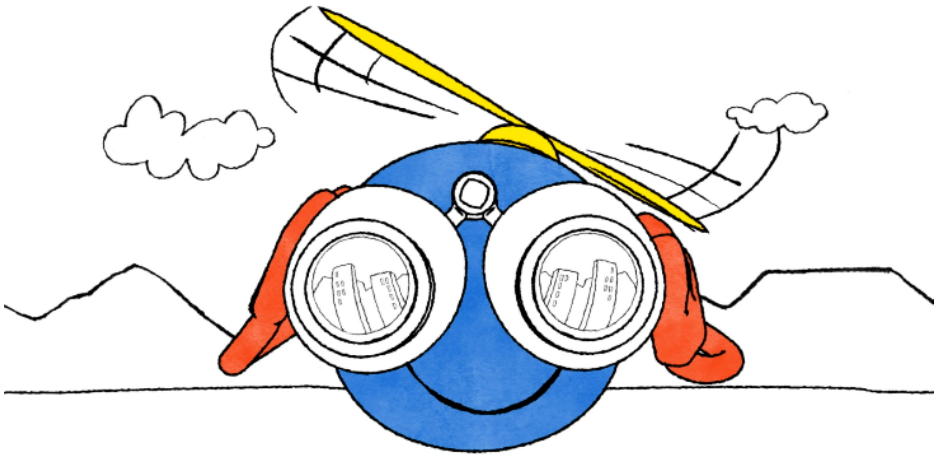
Let's rewind and uncover the origins of automation. Work has evolved throughout history, with people constantly improving their productivity using various tools and gadgets. However, let's be honest, work can still feel tedious at times, right?

Scan the QR code or watch this video to learn about the evolution of work.

From their humble beginnings, to modern marvels utilized in manufacturing and space exploration, robots have come a long way. With the advent of computers and technology, the need for software robots emerged, leading to the founding of UiPath in 2005.

As a major player in the automation industry, UiPath serves global customers and offers a range of capabilities to transform the way we work.

So, what can we do? What are our capabilities? Let's find out more!



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Automation categories

With UiPath, there are a bunch of automation categories that can help you get things done faster and easier!



Robotic Process Automation (RPA)

Through Robotic Process Automation (RPA), the creation of software robots that mimic human actions and effortlessly engage with digital systems becomes possible. RPA enables organizations to automate tasks across a diverse range of applications and systems, including legacy systems, web applications, and desktop applications.



Task Automation

Task automation targets repetitive and mundane tasks such as data entry, data extraction, file management, and document processing. By automating these routine activities, you not only save valuable time and resources but also minimize the risk of errors, allowing you to dedicate your focus to the most critical tasks.



Business Process Automation

UiPath empowers organizations to automate entire business processes by seamlessly connecting multiple tasks and systems. By implementing UiPath, your department or team can experience streamlined workflows, enhanced operational efficiency, and consistent execution of processes.



Test Automation

With Test Automation, organizations can automate the testing of their applications and systems, revolutionizing software quality. Automated testing speeds up release cycles and reduces the time and effort spent on manual testing.



AI and Machine Learning Integration

By integrating automation with AI and machine learning services, users can incorporate intelligent automation into their workflows. This includes leveraging natural language processing, digital image processing, and predictive analytics.



Process Mining and Discovery

Process mining and discovery tools empower organizations to identify and analyze their existing processes. This valuable insight leads organizations to the most promising opportunities for automation, enabling them to scale up their automation initiatives and prioritize projects with the greatest impact.

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Let’s test our knowledge!

1. Match the automation type with the most specific corresponding benefit.

Automation Type	Benefit
AI and Machine Learning Integration	Automates routine tasks like data entry
Robotic Process Automation (RPA)	Automates testing of applications and systems
Process Discovery Tools	Integrates AI into workflows
Task Automation	Identifies automation opportunities in processes
Test Automation	Creates software robots mimicking human actions

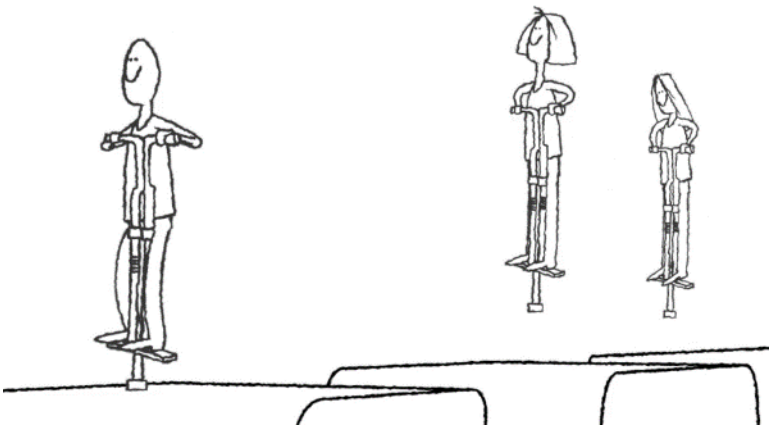
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Let’s playback the learning tape!

Automation transforms work by using robots to tackle repetitive tasks, allowing us to concentrate on high-value activities. With UiPath, you can deploy robots that automate daily tasks, organizational processes, and app testing. Enhance automation by integrating AI capabilities like language models and image processing. Leverage UiPath's process mining tools to extract insights from data, enabling better informed decision-making.

Here's a quick tip!

Practice — Make sure to practice and apply what you learn. Automation is a skill that requires practice to master, so we recommend you go through and practice the exercises provided in the course.



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Jot down your takeaways from this lesson…

Reflect on the following points and note down your thoughts.

- What did I learn and did not expect?
- What was difficult for me and how did I overcome it?
- What is the most useful thing that I learned in this lesson?

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How can automation help me?

You just got an overview of automation and all the different categories out there. But have you wondered what it can do for you? How can automation make your work a lot easier and more efficient? Or perhaps you're still wondering whether automation is appropriate for your organization and domain...



Well, you'll find all your answers here and more!

What can automation do to help me?

Job satisfaction

By automating repetitive tasks, you not only escape the burnout and boredom cycle but also gain precious time to dedicate to more engaging and fulfilling work.

Skill development

With automation handling mundane tasks, you have the perfect opportunity to embrace personal growth, learn new skills, and embrace higher-value roles within your organization.

Enhanced opportunities

Embracing automation and related technologies can greatly enhance your value in the job market, opening doors to exciting career opportunities and unlocking new pathways for professional growth.

Reduced stress

Letting automation handle mundane tasks gives you the freedom to embrace a healthier work-life balance, experiencing reduced stress and more time for what truly matters.

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How can automation help my organization?

Discover the top reasons why your organization simply can't afford to miss out on automation!



Saved cost

Automation brings valuable benefits to businesses by reducing labor costs and enabling teams to allocate their time and resources to more strategic and high-value activities.



Improved accuracy

With software robots, businesses can significantly reduce errors. This results in the generation of high-quality data, enabling informed decision-making and driving substantial improvements in overall business performance.



Increased efficiency

Robots excel at lightning-fast task execution, resulting in quicker turnaround times and enhanced output.



Enhanced compliance

By implementing standardized processes and generating automated audit trails, automation not only ensures adherence to regulations and internal policies but also offers a sigh of relief for your organization.



Improved scalability

One of the most impressive aspects of automation is its remarkable scalability, allowing organizations to effortlessly adapt to fluctuations in demand and navigate business needs with ease.



Optimizing resource allocation

Having their routine tasks automated, employees can focus on strategic, creative, and customer-centric endeavors. This unlocks higher productivity and fosters a culture of innovation, driving business success.

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What are some common automation use cases?

Automation has the power to tackle a wide array of repetitive tasks performed on computers, making it applicable across diverse industries and business units of a company. To get a better grasp of how automation can help, let's look at some examples that are common in so many lines of business.

Use Case	Challenge	Solution
Payroll processing 	• Payroll processing is a complex process action that companies take to pay their employees, keeping track of their presence, of their salaries, bonuses, and taxes. • It requires manual intervention month after month, every year.	• Robots here can extract the details that are required from handwritten time sheets, calculate the pay from their stipulated contracts, and pay them through banks.
Client information update 	• Every organization that has a CRM faces all sorts of related issues. It could be the client base spread across many geographies. • There are frequent calls to the back-end databases, and updates are coming from all sources.	• Robots can handle requests for updates in batches, reducing the load on back-end systems and improving performance and data quality.
Contract renewal process 	• The contract renewal process is complex in general. Not necessarily due to expectations and complications, but because of the number of operations and the synchronization between different departments and systems.	• Robots can handle the entire process starting with standardized communication with the client, processing the changes, drafting the documents, and updating the internal systems accordingly.
Financial statement reconciliation 	• The Financial Statement Reconciliation is a complex process. • The accounting teams and finance departments spend a lot of time matching orders, payments, losses, margins, and so on, with accounts and financial statements.	• It is a complex process, but robots can handle it from start to finish and ask for human validation when necessary.
Compliance reporting 	• As organizations grow, it becomes increasingly difficult to closely monitor the compliance requirements that each department must follow: - Reporting to the authorities. - Following internal procedures - Audit requirements - …and so, on	• Compliance reporting can be covered by robots with a low error rate and minimal human intervention.
Customer complaint processing 	• Customer complaints are always on the radar where their numbers and substance are an important indicator of business health and a good predictor of the future of the company.	• Customer complaints can be quickly categorized, and practical solutions are suggested using automation, ensuring 24/7 responses.

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Let’s test our knowledge!

1. **A colleague in your department doubts that automation can improve their workflow. Which three of the following use cases could be used as convincing examples to show that automation can be effectively applied to their tasks or processes?**

- ☐ Using automation for repetitive data entry tasks.
- ☐ Using automation to generate weekly reports.
- ☐ Using automation to create original and unique copy.
- ☐ Using automation for routine system backups.
- ☐ Using automation to negotiate complex contracts.



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Let’s playback the learning tape!

Automation can help businesses by increasing savings, accuracy, efficiency, and scalability. It can also improve compliance and resource allocation. As for you, automation can lead to increased job satisfaction and better opportunities, better work–life balance, and reduced stress!

Here's a quick tip!

Ask questions — don’t expect to get everything right the first time and don’t be afraid to communicate what you are unclear on. UiPath Academy has a large community of learners and experts who are willing to help.

If you need help with getting started, troubleshooting your automation, or just have some questions, UiPath Forum and UiPath Community are excellent resources with a large community of learners and experts you can reach out to! You can also reach out to the UiPath Support Team for assistance by submitting a ticket at https://www.uipath.com/support. We’re here to answer any of your inquiries!



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Jot down your takeaways from this lesson…

Reflect on the following points and note down your thoughts.

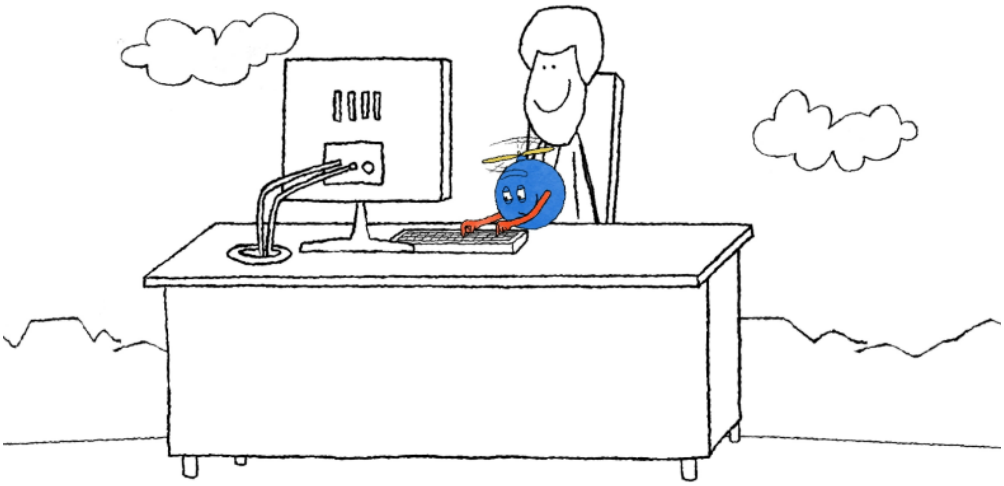
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Configure your first automation

It's time for your first attended automation project! Let's dive into the exciting world of UiPath Studio Web and configure a template project that will woo you!



Get ready for a thrilling experience!

In this brief exercise, you'll set up a ready-made automation using Studio Web. Studio Web is an intuitive, web-based platform that simplifies the creation and management of automation processes. It is designed to work on Windows, macOS, and Linux operating systems, making it a versatile solution for task automation across different platforms.

And the best part? No installation is needed!

The use-case context

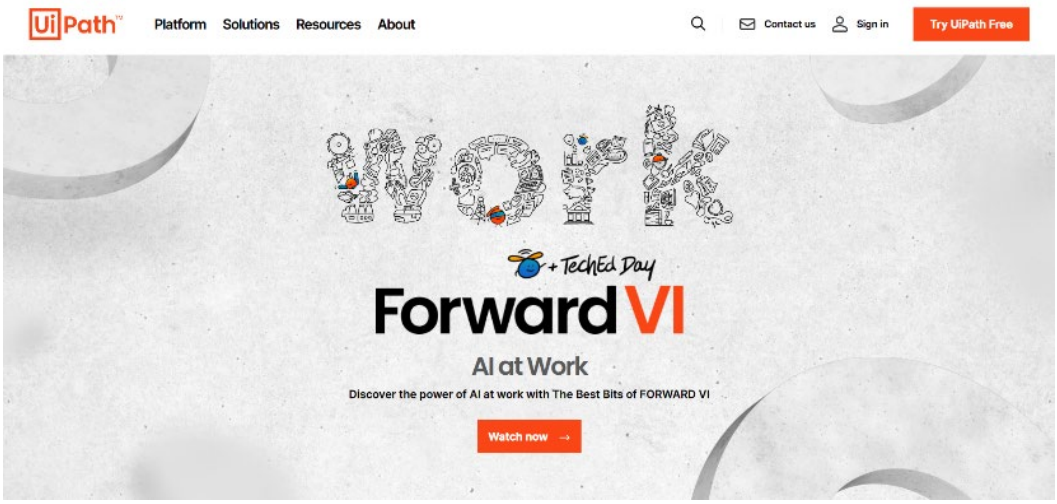
As a Sales Analyst, Taylor is responsible for tracking and analyzing Sales Opportunities to improve sales strategies and forecast future trends. Taylor has to manually extract data from a table called "Sales Opportunities" every day. Then they have to insert this data into a spreadsheet and email it to the management team. This process is tedious, time-consuming, and prone to human errors.

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How do I go about this?

All you need is an active UiPath account. Don't have an account yet? No problem! Follow these simple steps to unlock a world of automation possibilities.

1. To set up an UiPath account, visit https://www.uipath.com/ and click the **Try UiPath Free** button located at the upper right corner of the page.



2. Refer to the **Configure your first automation_Step-by-Step Instructions** PDF file for complete instructions on how to perform the automation.

Start automating!

It's time for you to put your knowledge into practice and create the automation. The primary objective is to make Taylor's daily task of extracting sales data and emailing it to the management team more manageable.

The project workflow has two parts:

1. Configuring a template.
2. Publishing the automation.

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Reflect on the following points and note down your thoughts.

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How do I team up with a robot?

We'd all like to have a helping hand to handle our most tedious tasks, don't we? Alright, we'd like to introduce you to that special someone, well, more like a bunch of them. In this section, we'll discover our automation robots and how they can work for you and with you!



How do I work with a robot?

A robot is a robot, right? In some ways, that's true. In our context…

A robot is a programmable machine that executes automations that people build.



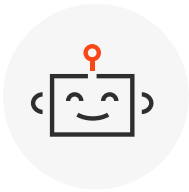
So, what is automation then?

An automation is a series of steps that tell the robot what to do.

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In UiPath, robots come in more than one flavor — attended and unattended. Let's look at them in detail.





Unattended robots

Attended robots **Unattended robots**

Unattended robots are those that don't need anyone to operate them. Here's a rundown of what they are.

	Unattended robots: What they are
What they do…	• Operate with minimal human intervention • Execute fully automated processes
Where they are deployed…	• Workstations • Servers • Mainframes • Web services
How they are triggered…	• Via an automation management tool (UiPath Orchestrator) • Based on a set schedule • By any activity or event • By another robot

Attended robots

Attended robots are those that need a little help from us humans to work. Here's a rundown of what they are.

	Attended robots: What they are
What they do…	• Operate on your computer • Become your trusted ally • Enhance both quality and productivity • Take charge of repetitive tasks • Enable you to accelerate processes promptly
Where they are deployed…	• On your PC • Laptop

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	• Departmental server
How they are triggered…	• By the user • By certain events on the user's machine

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Automation models

Business procedures can differ greatly depending on the industry. Some are straightforward, going from A to B, while others involve multiple interactions between people and systems to move forward. In some cases, a process might need to repeat certain steps to achieve the intended result. Basically, these business processes take time, focus, and energy, as their characteristics and intricacies change from one industry to another.

There are six models tailored to meet a broad spectrum of automation needs, merging the best of user-friendly and business-focused strategies.



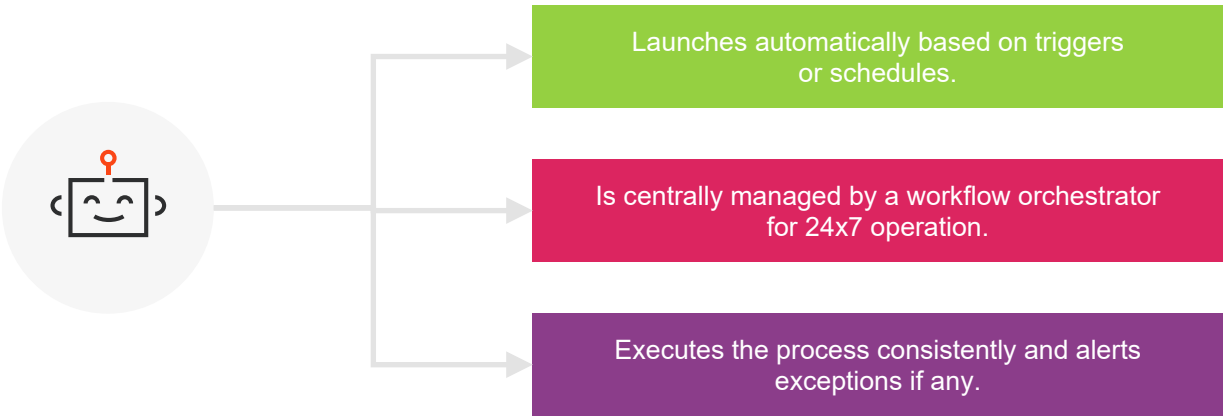
Let’s explore each of them in detail.

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1. Fully unattended

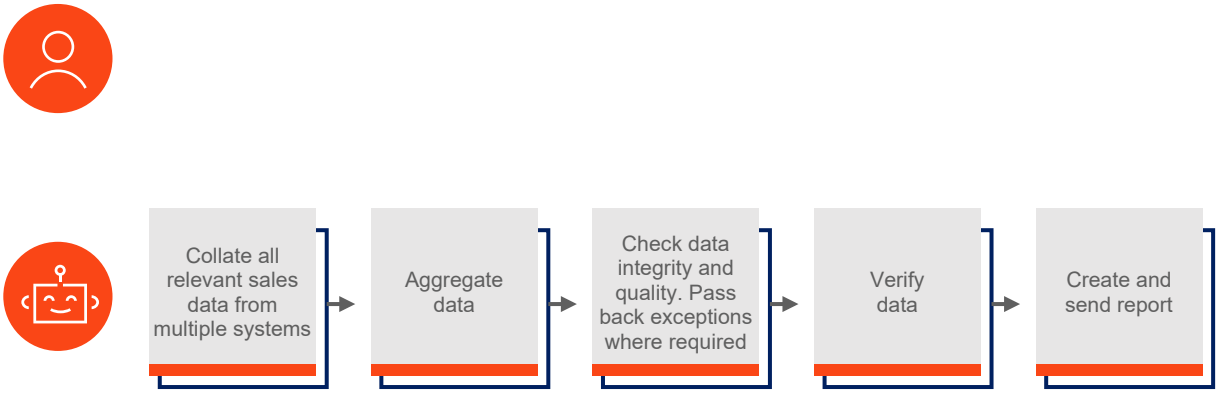
Automation is a game-changer for handling massive amounts of data. Moving and processing data manually across various systems is time-consuming and error prone. Fully Unattended robots are the perfect solution. They handle data processing without human intervention, ensuring efficiency and accuracy.

In this automation model, the unattended robot:



Let's consider a real-life scenario…

A global financial services company needs to collate sales figures from multiple systems, including ERP, CRM, and sales databases, every month. It's a labor-intensive task. Enter Fully Unattended automation! The unattended robot completes the entire process, improving accuracy and timeliness in generating the final report for senior management.

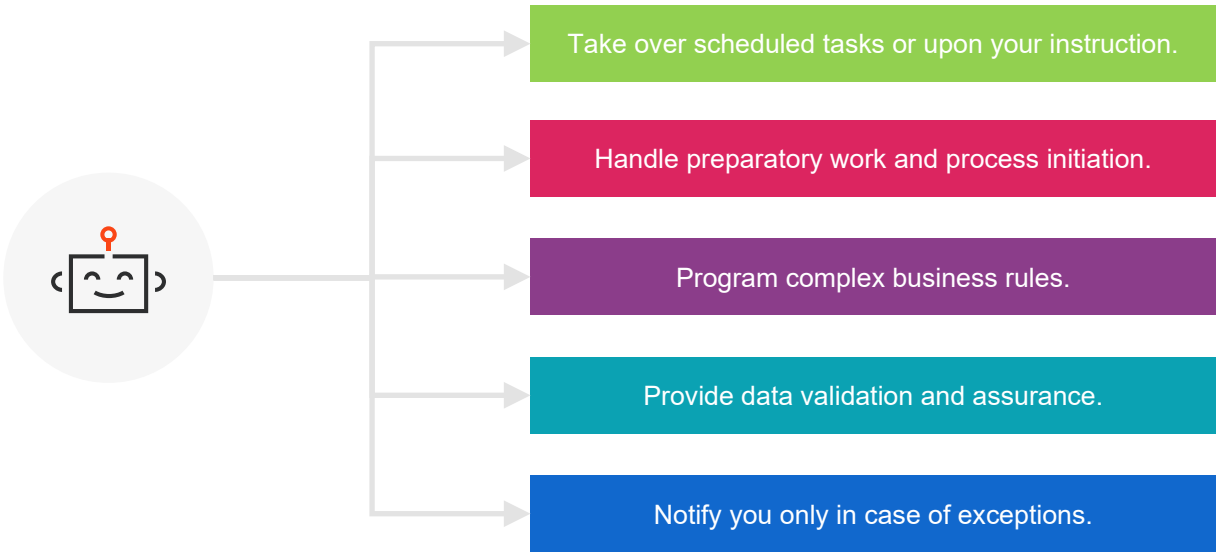


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2. Partially unattended

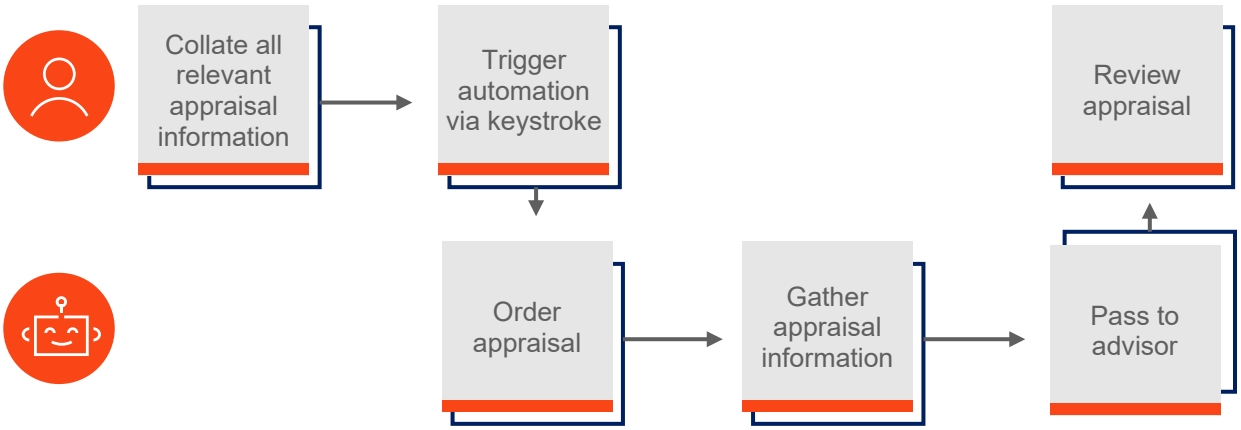
Tired of tedious and repetitive business processes? This automation model offers a solution! Start with human involvement and then hand off the rest to a Partially Unattended robot. This frees you to focus on high-value activities while ensuring the process is completed successfully with thorough documentation and notifications.

In this automation model, the partially unattended robot can:



Consider a real-life example, mortgage appraisals…

Ordering them manually is time-consuming and prone to mistakes. However, with Partially Unattended automation, lenders initiate the process, and the robot takes over. Appraisal time is significantly reduced, leading to cost savings and fewer mortgage escalations.

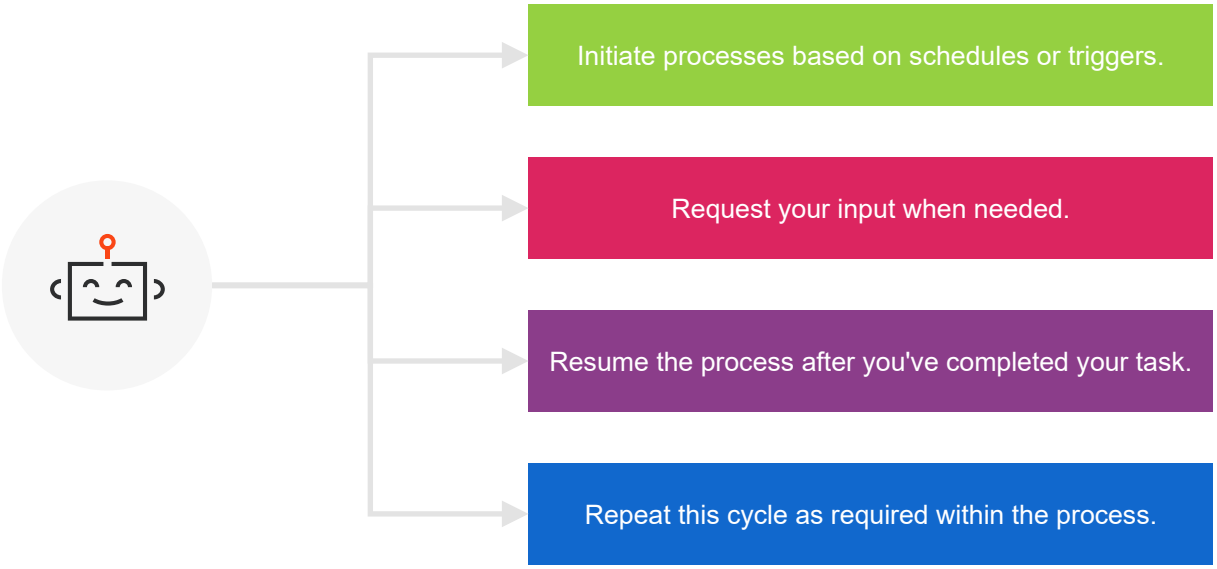


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3. Human in the loop

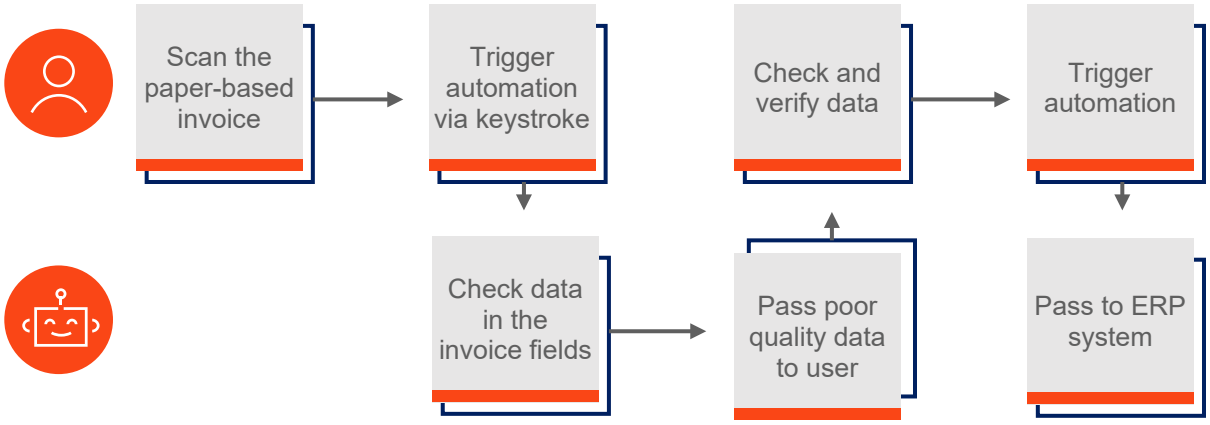
Next in the row is the human in the loop automation model. Certain process types requiring intermittent human intervention and complex decision-making can be challenging to automate with traditional tools. However, with this automation model, business rules are embedded into the process, allowing the robot to perform tasks and prompt for input only when necessary, simplifying your role.

In this model, the unattended robot can:



Let's take for example an invoice processing use-case…

In invoice processing, OCR technology is used to capture information from scanned invoices. However, if the data quality is low, automation alerts you to review and validate the information before entering it into the ERP system. By combining RPA with AI, the system learns from your validations, reducing the need for future intervention.

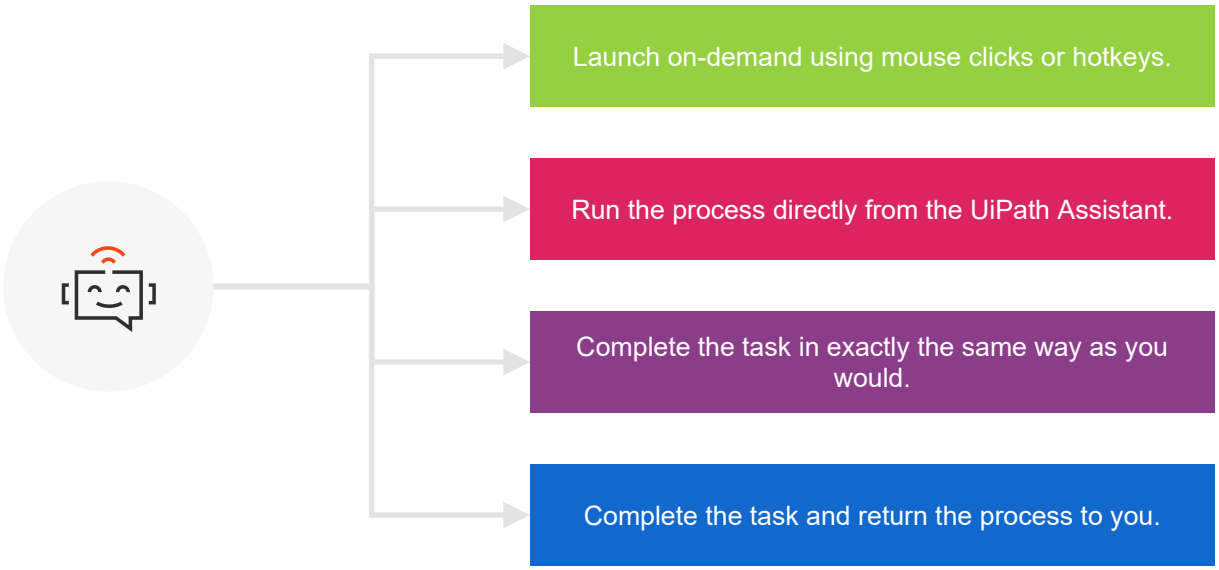


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4. Attended, interval

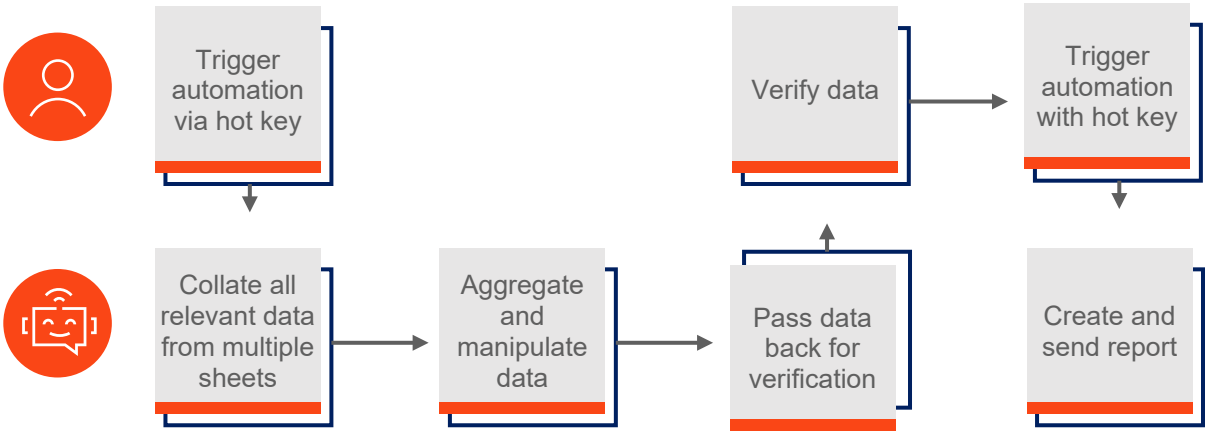
In the attended, interval automation model, the attended robot can launch on-demand, run processes from the robot tray, and mimic your actions precisely. The only drawback is that sometimes your machine is occupied during the robot's task. However, you can use the free time for offline tasks.

In this model, the attended robot can:



Let's consider a real-life banking scenario…

Debit cards at a bank may be declined for 81 reasons. Smooth customer experience relies on efficient call handling by our staff, explaining the reason. Previously, staff faced a cumbersome process of navigating screens, writing down, and calculating data to provide answers. Now, an attended robot swiftly performs these tasks, instantly presenting employees with a clear description of the declined card's reason.

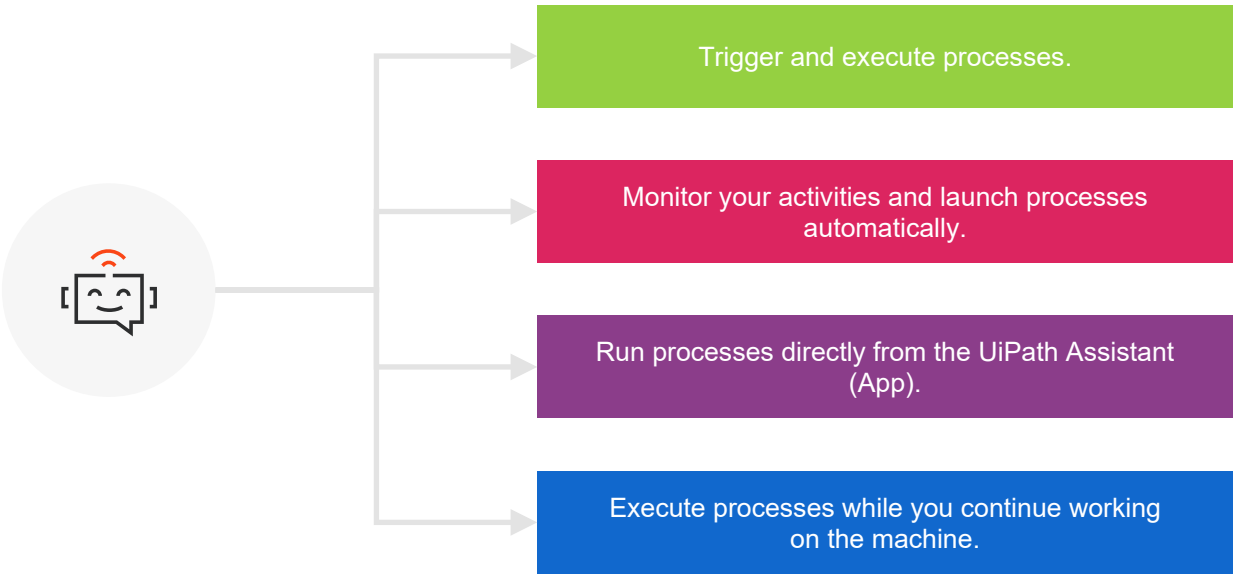


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5. Attended in tandem

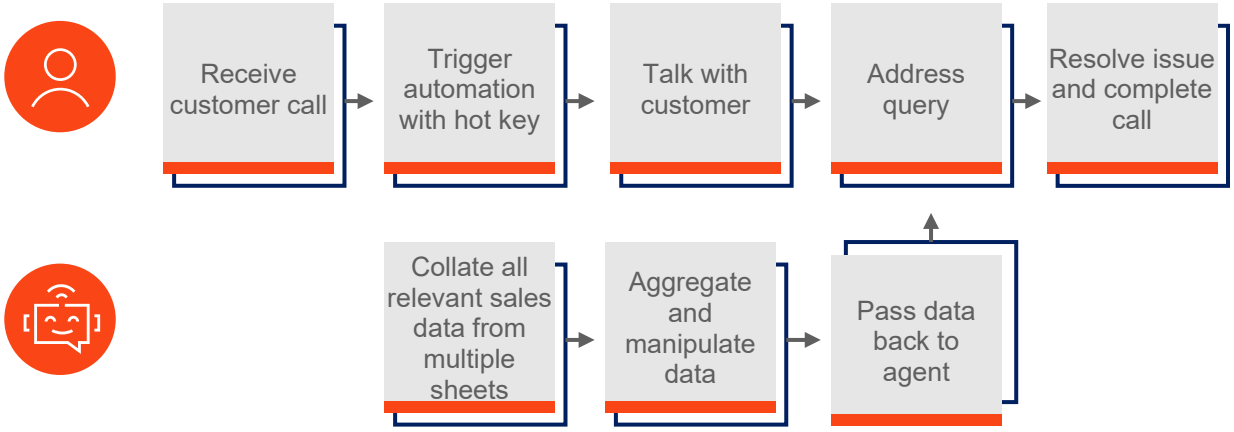
Then you have the attended in tandem automation model. In certain situations, it may not be practical or efficient for a robot to take control of your machine. Instead, consider the Attended in Tandem automation model. This approach allows the robot to work alongside you, providing valuable information while you focus on your primary tasks.

In this model, your attended robot can:



In a customer service scenario…

Customer service agents often face the challenge of navigating multiple systems during customer calls. This process can be time-consuming and prone to errors, leading to frustrations for both the agent and the customer. With Attended in Tandem automation, agents can seamlessly handle customer queries while the robot operates in the background. The robot accesses, compiles, and presents the necessary information, enabling agents to address customer needs more efficiently. This boosts productivity, reduces resolution times, and ultimately enhances customer satisfaction.

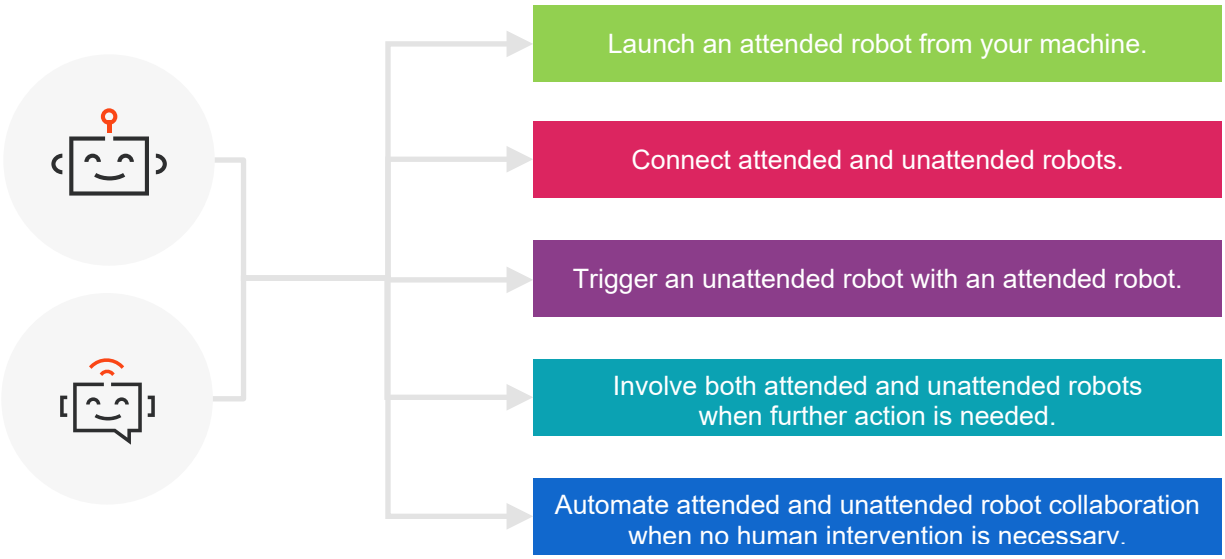


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6. Hybrid automation

Last in the row is the hybrid automation model. Hybrid automation merges attended and unattended robots within a unified platform. Attended robots assist with customer-facing tasks, while unattended robots handle heavy back-end processing. It automates end-to-end business processes, providing flexibility and scalability.

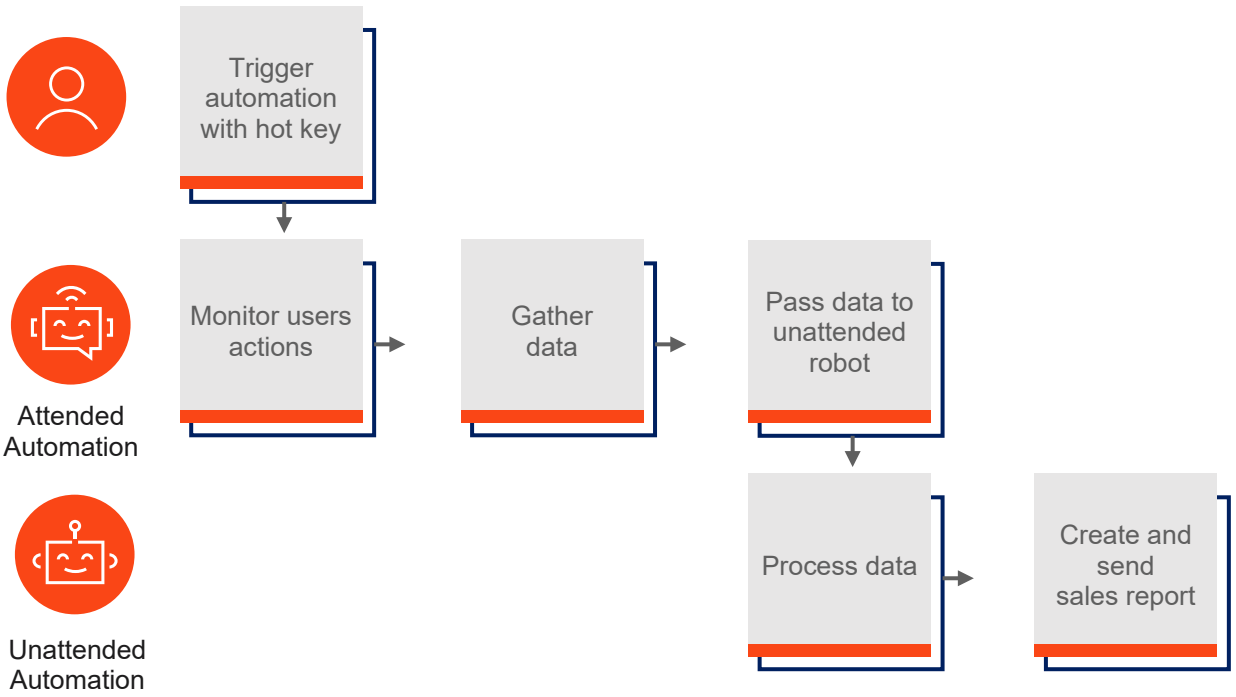
In the hybrid automation model, you can:



Let's take this example…

Sales teams can leverage UiPath's Hybrid automation for improved productivity. Attended automation monitor activities, capturing relevant data. At regular intervals or in real-time, the attended robot transfers information to an unattended robot for further processing and sales report generation.

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Let's test our knowledge!

1. Which of the following three statements accurately describe the characteristics of attended and unattended automation/robots?

- ☐ Unattended automation requires little to no human intervention.
- ☐ Unattended robots handle the routine tasks involved in a process.
- ☐ In Attended automation, robots work alongside you.
- ☐ Attended robots handle high-volume transaction-based processes.
- ☐ Attended robots reside on your PC, laptop, and departmental server.
- ☐ Unattended robots reside on your PC, laptop, and departmental server.



2. Which of the following automation models allow the robot to handle tasks in the background while the user continues working?

- ○ Human in the Loop
- ○ Attended in Tandem
- ○ Fully Unattended
- ○ Hybrid automation



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Let’s playback the learning tape!

Attended bots operate on a user's computer, helping them with tedious tasks and speeding up certain processes. Unattended bots perform fully automated processes with little to no human interference. UiPath offers six automation models to accommodate various processes and industries, differing mainly in levels of human-bot engagement.

Here's a quick tip!

Explore — To get started, focus on understanding the core elements and functions of UiPath. This will set you up for success and help you grasp the different automation models available, as well as their workings. And don't forget to tap into UiPath's rich, online resources, at UiPath Academy. With UiPath, sky's the limit, and your learning experience is about to get a whole lot more exciting!



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Jot down your takeaways from this lesson…

Reflect on the following points and note down your thoughts.

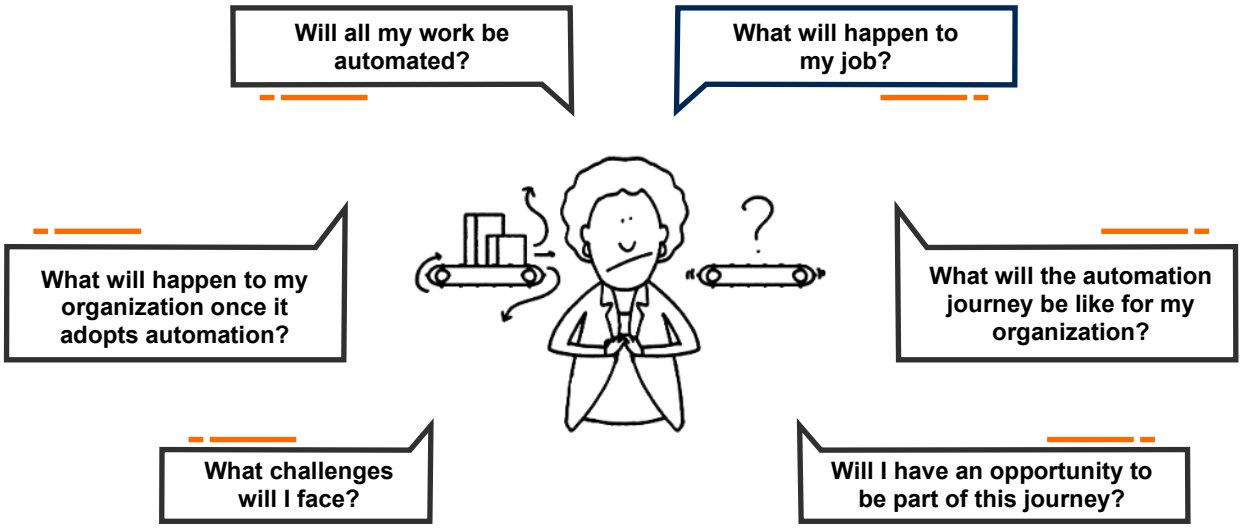
- What did I learn and did not expect?
- What was difficult for me and how did I overcome it?
- What is the most useful thing that I learned in this lesson?

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How can automation fit into my business?

Robots are not job-stealers! In fact, they are here to help us unlock new opportunities and take our skills to greater heights. With automation at our fingertips, we have the power to achieve incredible success like never before!

You may have questions, like …



These are some of the questions that can pop up in every employee's mind when first hearing about automation, just like Arden, a Customer Service Executive for a home appliance company, XYZ Inc. But in a short time, Arden turned from an automation nonbeliever to an Automation Champion!

Their transformation began when their department was selected for the automation pilot project. Initially, the team experienced a mix of hesitation and curiosity, which is expected with major changes. However, as you'll see, that changed quickly.

Let’s take a peep into the evolution of XYZ Inc. as seen through the eyes of Arden!

Scan the QR code or watch this video to learn about the evolution of XYZ Inc.

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The different roles that automation can create

Automation is revolutionizing the way businesses operate, streamlining their processes and freeing up vital resources. Check out this collection of new roles that automation can create within a company!

Automation Citizen Developer	Business Analyst	Test Automation Manager
Communication Mining Model Trainer	Automation User	Project Manager
Automation Developer		Infrastructure Engineer
Process Mining Analyst		Automation Developer – DU Specialist
Solution Architect	Test Automation Engineer	Process Mining Developer

Refer to the Role-based Journeys.zip file to find out about the different roles that automation can open up in your organization.

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Let’s test our knowledge!

1. **When an organization is in the initial stages of its automation journey, who raises automation requests?**

- ○ Automation Users
- ○ Business
- ○ The Automation CoE
- ○ Senior Management



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Let’s playback the learning tape!

Contrary to common misconceptions, automation and digital transformation open up opportunities for skill advancement and career success. As automation is integrated throughout the organization, new roles and possibilities emerge, fostering professional growth and encouraging employees to expand their expertise.

Here's a quick tip!

Start slow and steady — To get started, try implementing automation in small steps. Begin by automating a few simple tasks or processes and gradually increase the complexity of the tasks that you automate. When introducing automation to your business, focus on the benefits it will bring, such as freeing up time to focus on more important tasks, etc. Automation can be an exciting way to help your business grow and increase efficiency—if you do it right!



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Jot down your takeaways from this lesson…

Reflect on the following points and note down your thoughts.

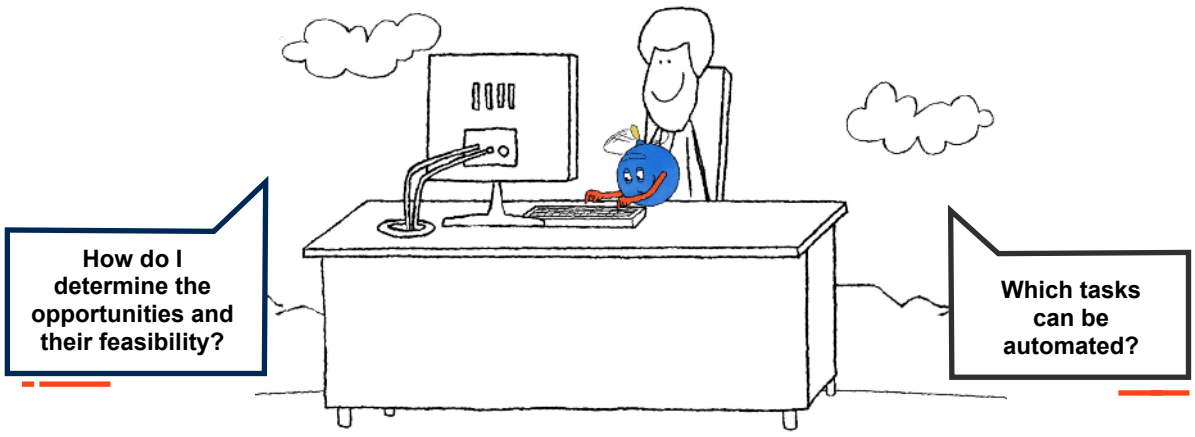
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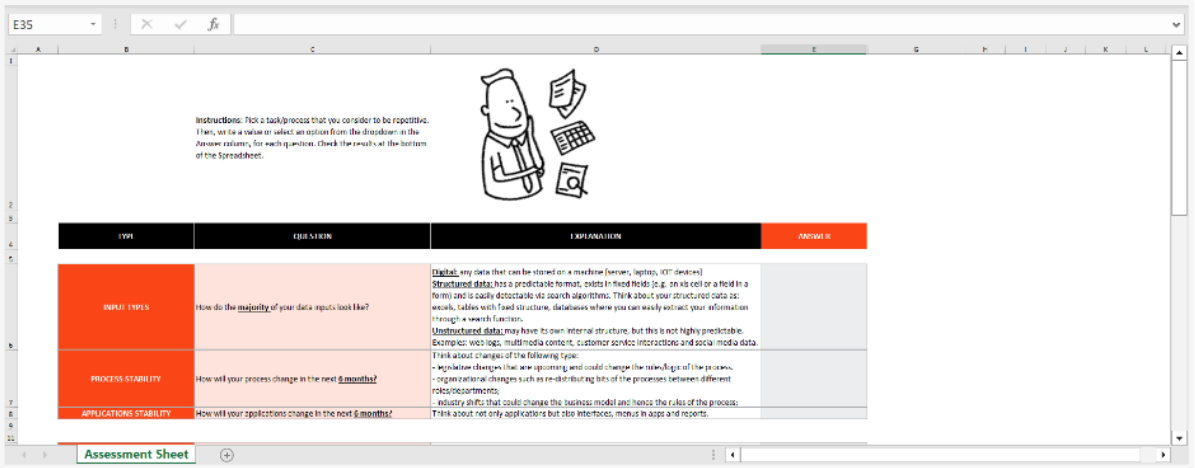
How do I identify automation opportunities?

By now, you should understand that automating tedious, repetitive tasks frees up your time. However, you might wonder which tasks can be automated, their feasibility, and how to determine that. Well, we have great news for you! Our Process Assessment tool can help you identify **which tasks can be automated, their feasibility, and how to submit your automation ideas**.

So, get ready to say goodbye to mundane tasks and hello to a life filled with excitement and productivity!



The Process Assessment tool



The Process Assessment tool helps you determine if your automation idea is a good fit for automation and to evaluate its feasibility. This tool comprises a set of questions that assess the process's characteristics and current state. Your answers to these questions will shape the outcome of the assessment. The result of the assessment provides a clear indication of the benefits of automating the specific process, the simplicity of its implementation and development, as well as the potential hours you could save

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Here's an example…

Taylor is an accountant. One of their major tasks is processing invoices. This task is manual and time consuming. Taylor's company is planning to take this task as a pilot as they are getting started on their automation journey.

Here are the steps involved in the process.

01 Receive invoices via email → **02** Download all the attachments → **03** Check for discounts on the portal → **04** Save as PDFs and email

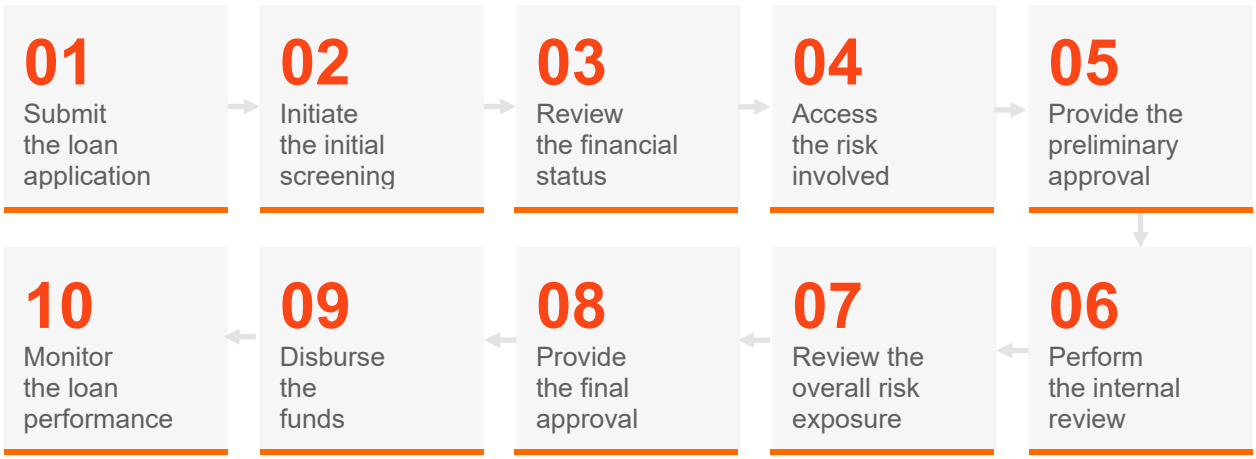
Scan the QR code or watch this video to learn the assessment results of the invoicing processing.

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Here's another example…

You've got the hang of how the Process Assessment tool works, right? To deepen your understanding, we have another example for you. Taylor's boss has now asked them to evaluate the commercial loan application process for a banking client. In this use case, we'll analyze a commercial loan application process. Let's see what the tool reveals about this one.

Here are the steps involved in the process.



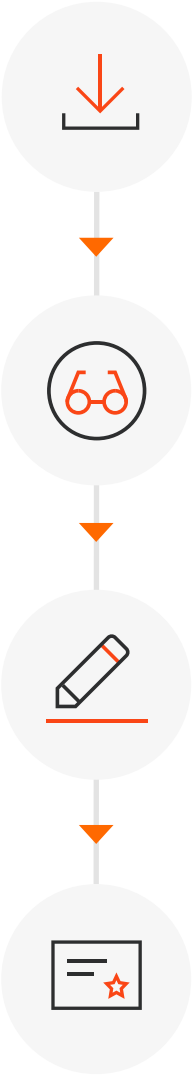
Scan the QR code or watch this video to learn the assessment results of the loan application process.

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It's your turn now!

Are you ready for an exciting activity? Get ready to witness the power of the Process Assessment tool firsthand! We have an exciting use case lined up for you, so buckle up and let's dive into the world of process assessment!

First things first, here are the steps you need to take to complete your assessment…



1. Download and open the Process Assessment Tool document. Refer to the **Process Assessment Tool – Introduction to Automation** zip file to download the tool. If you aren't a Microsoft Office user, you can convert it to a format that fits your environment (e.g., upload it to your Google Drive and convert it to a Google Sheets document).

2. Read the use case in the student guide carefully. It's important to have the use case figured out from the beginning to have a clear picture of what you're working with. Feel free to take notes on the steps of the process (here, we're lending you a hand!) and other details you find noteworthy.

3. Fill in your answers in the Process Assessment tool. There's a dedicated column where you can fill in the answers to the questions in the Process Assessment tool. With this step, you'll showcase if and how much the use case meets the set automation criteria. For some of the criteria, you can consult the useful tips in the "Explanation" column. It's also important to note that, depending on each assessor's perspective on the use case, the solution may slightly vary. Try to be as thorough as possible and take into account the small details, like sub-steps that may be involved.

4. Check out the Automation Score based on the process complexity, benefits of automation, and ease of implementation! At the end of the document, after filling in your answers, you'll see the evaluation results. Check yours against the question at the end of this lesson and, if you're curious, against our version of it.

Use the Process Assessment tool in Jamie's use case to evaluate the automation potential of their process. The details of the process that you need to assess are in the form of an interview. Jamie answers the questions they got from an analyst to shed some light on the situation.

Go through each part of the interview and answer the questions in the assessment tool based on the provided information!

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The use case…



Hi Jamie! First things first, tell me a bit about you, your role, and your challenge.

Hello! Sure. I'm Jamie, a Human Resources specialist in a fast-growing tech company, and I'm responsible for conducting the onboarding process for new hires.



Currently, the company is exploring automation opportunities to free up employees from monotonous, time-consuming tasks. For starters, they are choosing a few processes to automate and conduct a pilot. My challenge here is to write up a proposal explaining how automating the onboarding process can benefit both the HR team and the organization. Here's where I need some help figuring out how feasible this would be.



Great! This is where the Process Assessment tool comes in. Before jumping into it though, please describe briefly what the onboarding process is like.

First, an HR specialist receives a new employee's information via email, in the form of a standardized document.

Then, they create a new employee account in the internal system of the company. At this point, they also set up the appropriate access of the new hire, specified in the document with the employee's details.



Next, the HR specialist prepares a welcome package with essential HR documents. They upload the documents in the "Onboarding" section of the employee portal and send them to the new employee via email as well, accompanied by a welcome message.

And, last but not least, they schedule orientation sessions with various team members



Thanks! Now, let's dive in. What's the majority of your input data like?

Well, everything we do is digital, that's for sure! Otherwise, the employee data we work with is all standardized, even though it reaches us through the email. It's held in documents with a set structure. So are the documents in the HR kits and the scheduling of the orientation sessions. We do that with the help of a form.



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How stable is your process and the infrastructure you're working with?

As the procedures are clear and effective, we expect no changes in the next year. Not in the process itself, nor in the applications that we use





Great! Let's talk a bit about benefits now. How many of your team members perform this process?

There are two of us who conduct the onboarding process at the moment.





How often does the process take place? Can you think of a frequency and a volume here?

Of course! We go through onboarding quarterly, as different business units expand at different times and in the Summer we welcome our interns. With each time, we expect a number of approximately 25 people to be onboarded.





And how long does it take you to complete the onboarding process for a new hire?

I'd say about an hour.





What can you tell me about the number of human errors?

Well, there are quite a few steps involved in this process. And we work with multiple apps and some variables. So, errors happen naturally. We try our best, but on average, we need to go back and revise something for 2 out of 10 people. It might be something in the HR kit, some scheduling error, or something else. Many things can go wrong.
.



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I get it. It sounds like a rather complex process. Let's think about its consistency. Does the process have any peaks we should be aware of? Is there any unpredictability?



I'd say the predictability is high, as we have set periods of recruitment for different teams. When those end, the onboarding starts, near or at the end of the quarter



Got it! Now let's see how complex the process really is. First, how many steps are there altogether?

Alright! Well, after we get the employee's details attached via mail, we have to first download the document. Then, we create the employee profile, take the info from the details document and set up the profile. Based on the employment type, business unit, and team, we also set the permissions in the system. Up until now, we have 5 steps.



Next, based on the same criteria I talked about above, we check against a checklist of materials and generate the HR kit: we duplicate specific documents and upload them in the employee portal. We also send them via email to the new hire. For the email, we customize a pre-existing template as a welcome message. And now we're at 10 steps!

For the orientation sessions, for each role we have a list of people that we contact, asking to fill out a form with their availability. Then, we take the results and schedule the meetings in the calendar. These are another 3 steps, but that's it!



How can you describe the difficulty of the decisions involved in the process and what percentage of cases are you typically unable to complete?



The decisions involved are quite simple: we have a set of criteria that we use and we need to answer the same question with "yes" or "no": "Does this criterion apply?"

And since I've been responsible for this process, there wasn't any case where we weren't unable to complete it.



With regards to applications that you use, how many are there? And do you happen to use virtual machines?



We don't need virtual machines, no. I'd say 4-5 apps.

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Awesome, thanks! Now let's go back a bit to the data you use. You said all of your data is digital, no exceptions?

That is correct, yes.

Okay. How about its nature? How much of it is scanned and how much of it is structured?

We don't use scanned documents and I'm confident to say that over 90% of the data is structured.

Great! Thanks for this awesome interview. We got valuable info from you and we've got everything we need to complete the process assessment!

Alright! Now that you've gathered your inputs, get started with populating the Process Assessment tool. Once you're done, compare your solution to the one found in the **Process Assessment Tool – Onboarding Process – Solution** Excel file for a potential solution.



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Let’s playback the learning tape!

The Process Assessment tool makes it easy to determine which processes can be automated by asking a series of questions and calculating a score. The assessment result highlights benefits, suitability, ease of development, annual liberated man-hours, and the estimated error reduction percentage during automation.

Here's a quick tip!

Dive Deep — When using the Process Assessment tool, make sure to thoroughly analyze the process you want to automate in order to obtain an accurate assessment. Be sure to include as much detail as possible in the assessment, such as the number of steps involved, the complexity of the process, and the amount of data that needs to be processed. Also, be sure to consider any potential risks or constraints that could affect the automation of the process.



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Jot down your takeaways from this lesson…

Reflect on the following points and note down your thoughts.

- What did I learn and did not expect?
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Final quiz



1. What role did Arden play in XYZ Inc.'s Change Management department after becoming an Automation Champion?

- ○ Building software robots.
- ○ Encouraging and supporting the adoption of automation.
- ○ Implementing automation projects.
- ○ Conducting Process Mining analysis.

2. A customer service agent needs real-time assistance with accessing and presenting necessary information while handling a customer call. Which automation model is best suited for this situation?

- ○ Attended in tandem.
- ○ Fully unattended.
- ○ Human in the loop.
- ○ Partially unattended.

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3. **Per the scenario presented in the course, which four of the following steps are part of the process of setting up an automation in Studio Web?**

- ☐ Configure your connections.
- ☐ Publish the automation.
- ☐ Test the automation by running it.
- ☐ Obtain peer approval before running the automation.
- ☐ Select a template.
- ☐ Set a password before running the automation.

4. **In a fully unattended automation scenario, what is the primary characteristic of the unattended robot?**

- ○ It operates without human intervention.
- ○ It has limited capabilities in handling multiple systems.
- ○ It needs regular input from users.
- ○ It only works during business hours.
- ○ It requires constant human supervision.

5. **In the Invoice Process scenario, how does the Process Assessment tool determine if Taylor's invoice processing task is a good candidate for automation?**

- ○ The tool relies solely on structured input data percentage.
- ○ The tool checks the total number of transactions per day.
- ○ The tool focuses on the number of virtual machines used in the process.
- ○ The tool calculates an automation score considering benefits, ease of implementation, and potential error reduction.
- ○ The tool evaluates the process based on its complexity.

6. **What types of robots are involved in hybrid automation?**

- ○ Unattended robots only.
- ○ Attended robots only.
- ○ Attended and unattended robots.

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7. What is the main goal of automation?

- ○ To demand more human intervention in business processes.
- ○ To increase task/process complexity.
- ○ To boost efficiency, accuracy, and productivity.
- ○ To replace humans in all of the work they have to do.

8. What are two main advantages that automation brings?

- ☐ It identifies and fixes vulnerabilities in company processes.
- ☐ It reduces errors for better data quality and improves compliance.
- ☐ It helps streamline standardized processes and increases scalability.
- ☐ It makes decisions on its own, autonomously, and independently from any human input.
- ☐ It increases the workload of human employees.

9. Why is it necessary to establish third-party connections for some automations in UiPath Studio Web?

- ○ To allow the automation to run faster and more efficiently.
- ○ To facilitate automatic updates to the automation workflow.
- ○ To ensure secure storage of sensitive information within the project files.
- ○ To enable the automation to access and interact with external services (such as the Google Workspace Applications).

10. In what two ways should a task or process be like to be automatable?

- ☐ With a high degree of complexity, involving multiple decision-makers.
- ☐ Requiring creative problem solving.
- ☐ Constantly changing.
- ☐ Repetitive and predictable.
- ☐ Rule-based.

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Knowledge checks: correct answers

Match the automation type with the most specific corresponding benefit?



Automation Type	Property
AI and Machine Learning Integration	Integrates AI into workflows
Robotic Process Automation (RPA)	Creates software robots mimicking human actions
Process Discovery Tools	Identifies automation opportunities in processes
Task Automation	Automates routine tasks like data entry
Test Automation	Automates testing of applications and systems

A colleague in your department doubts that automation can improve their workflow. Which three of the following use cases could be used as convincing examples to show that automation can be effectively applied to their tasks or processes?

- ☐ **Using automation for repetitive data entry tasks.**
- ☐ **Using automation to generate weekly reports.**
- ☐ Using automation to create original and unique copy.
- ☐ **Using automation for routine system backups.**
- ☐ Using automation to negotiate complex contracts.



These tasks are ideal for automation due to their repetitive, rule-based, or time-consuming nature. Automation can efficiently handle data entry, reducing errors and allowing employees to focus on higher-value tasks; streamline report generation by gathering, processing, and presenting data consistently; and manage routine system backups, ensuring data security and redundancy without human intervention.

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Which three of the following statements accurately describe the characteristics of attended and unattended automation/robots?

- ☐ **Unattended automation requires little to no human intervention.**
- ☐ Unattended robots handle the routine tasks involved in a process.
- ☐ **In Attended automation, robots work alongside you.**
- ☐ Attended robots handle high-volume transaction-based processes.
- ☐ **Attended robots reside on your PC, laptop, and departmental server.**
- ☐ Unattended robots reside on your PC, laptop, and departmental server.



Unattended automation requires little to no human intervention while the robots handle high-volume transaction-based processes. The unattended robots reside on a dedicated workstation, server, mainframe, or web service. In attended automation, robots work alongside you on routine tasks involved in a process. They reside on your PC, laptop, and departmental server.

Which of the following automation models allows the robot to handle tasks in the background while the user continues working?

- ○ Human in the Loop.
- ○ **Attended in Tandem**.
- ○ Fully Unattended.
- ○ Hybrid automation.



In an Attended in Tandem model, automation works alongside users, allowing them to work on higher-value tasks while the bot handles repetitive tasks in the background.

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When an organization is in the initial stages of its automation journey, who raises automation requests?

- ○ Automation Users
- ○ Business
- ○ **The Automation CoE**
- ○ Senior Management



In the initial stages of an organization's automation journey, the Automation Center of Excellence (CoE) typically raises automation requests. This is because the Automation CoE is responsible for overseeing the organization's automation strategy, including identifying, prioritizing, and implementing automation opportunities. It ensures that the right processes are selected for automation, in line with the organization's objectives and goals, while also maintaining standards and best practices.

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Final quiz: correct answers

1. **What role did Arden play in XYZ Inc.'s Change Management department after becoming an Automation Champion?**

- ○ Building software robots.
- ○ **Encouraging and supporting the adoption of automation.**
- ○ Implementing automation projects.
- ○ Conducting Process Mining analysis.

As an Automation Champion, Arden's role was to encourage and support his colleagues in the adoption of automation within the department. By sharing his experiences and helping others embrace the benefits of automation, Arden played an essential part in fostering an automation-centric mindset and furthering the success of automation initiatives within the organization.

2. **A customer service agent needs real-time assistance with accessing and presenting necessary information while handling a customer call. Which automation model is best suited for this situation?**

- ○ **Attended in tandem.**
- ○ Fully unattended.
- ○ Human in the loop.
- ○ Partially unattended.

Attended in Tandem model supports users in real-time while they perform their tasks. In the given scenario, the customer service agent requires support in accessing and presenting data during a customer call. With an attended robot operating in tandem, the robot seamlessly works in the background, gathering and presenting the necessary information to the agent while the agent focuses on addressing the customer needs.

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3. Per the scenario presented in the course, which four of the following steps are part of the process of setting up an automation in Studio Web?

- ☐ **Configure your connections.**
- ☐ **Publish the automation**.
- ☐ **Test the automation by running it.**
- ☐ Obtain peer approval before running the automation.
- ☐ **Select a template.**
- ☐ Set a password before running the automation.

In the given scenario, the automation requires connections to Google Drive, Google Sheets, and Gmail, which are external third-party services. Installing a browser extension allows UiPath to communicate and work efficiently within browser tabs. The automation sends an email as part of the overall process. Configuring the email address is necessary to ensure the automation can access and use the correct account for sending emails. Once the automation is set up and tested, publishing it helps make it easily accessible for future use. You can run it anytime you want; it doesn't require setting up a password or any peer approval.

4. In a fully unattended automation scenario, what is the primary characteristic of the unattended robot?

- ○ **It operates without human intervention.**
- ○ It has limited capabilities in handling multiple systems.
- ○ It needs regular input from users.
- ○ It only works during business hours.
- ○ It requires constant human supervision.

Fully unattended robots are designed to autonomously perform tasks within an automation process. For example, in the global financial services company scenario, the unattended robot completes the entire data collation process from multiple systems utilizing its pre-programmed instructions. It executes the process consistently and efficiently, freeing up human resources for more value-added tasks and minimizing the risk of manual errors. The robot alerts humans only when exceptions occur, thereby demonstrating its ability to operate without the need for constant human supervision or intervention.

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5. In the Invoice Process scenario, how does the Process Assessment tool determine if Taylor's invoice processing task is a good candidate for automation?

- ○ The tool relies solely on structured input data percentage.
- ○ The tool checks the total number of transactions per day.
- ○ The tool focuses on the number of virtual machines used in the process.
- ○ **The tool calculates an automation score considering benefits, ease of implementation, and potential error reduction.**
- ○ The tool evaluates the process based on its complexity.

The Process Assessment takes into account factors such as the benefits of automation (e.g., time savings, error reduction), the ease of development and implementation, and how much the process is expected to change.

By analyzing these factors and calculating an automation score, the Process Assessment tool provides a comprehensive assessment of the feasibility and potential impact of automation. This helps Taylor and the company make informed decisions about whether to proceed with automating the invoice processing task and how it might benefit employees and the organization.

6. What types of robots are involved in hybrid automation?

- ○ Unattended robots only.
- ○ Attended robots only.
- ○ **Attended and unattended robots.**

Hybrid automation refers to a combination of both attended and unattended robots working together within a unified platform. In this approach, attended robots typically assist with user-facing tasks, while unattended robots focus on handling heavy back-end processing. By combining the capabilities of both robot types, hybrid automation can automate end-to-end business processes more effectively.

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7. What is the main goal of automation?

- ○ To demand more human intervention in business processes.
- ○ To make increase task/process complexity.
- ○ **To boost efficiency, accuracy, and productivity.**
- ○ To replace humans in all of the work they have to do.

The primary objective of automation is to improve work processes by reducing the need for manual labor, especially for repetitive tasks. This leads to increased efficiency, as tasks can be performed more quickly and accurately by automation tools. Nevertheless, let's not forget that robots, while they mimic human behavior, aren't capable of everything we, humans, are.

8. What are two main advantages that automation brings?

- ☐ It identifies and fixes vulnerabilities in company processes.
- ☐ **It reduces errors for better data quality and improves compliance.**
- ☐ **It helps streamline standardized processes and increases scalability.**
- ☐ It makes decisions on its own, autonomously, and independently from any human input.
- ☐ It increases the workload of human employees.

Automation streamlines business processes through software robots, which execute tasks efficiently and consistently. This leads to improved process management and operational efficiency. By automating tasks, businesses minimize human error, enhance data quality, and free up employees of repetitive work. However, robots are not humans. They execute automation workflows with clear instructions, built by humans. They cannot make decisions independently, outside of their instructions' scope.

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9. Why is it necessary to establish third-party connections for some automations in UiPath Studio Web?

- ○ To allow the automation to run faster and more efficiently.
- ○ To facilitate automatic updates to the automation workflow.
- ○ To ensure secure storage of sensitive information within the project files.
- ○ **To enable the automation to access and interact with external services (such as the Google Workspace applications).**

By establishing connections with these third-party services, the automation can perform the intended tasks accurately, which is essential for a successful automation run. These connections enable the automation to perform tasks across multiple applications, facilitating seamless data exchange and process execution between different platforms.

10. In what two ways should a task or process be like to be automatable?

- ☐ With a high degree of complexity, involving multiple decision-makers.
- ☐ Requiring creative problem solving.
- ☐ Constantly changing.
- ☐ **Repetitive and predictable.**
- ☐ **Rule-based.**

Tasks and processes requiring complex decision-making and creativity, with a high degree of unpredictability or that are constantly evolving, are best suited for people to handle. The robots are able to pick up whatever tedious, repetitive work remains! They aren't capable of "thinking" on their own, so judgment calls or working in an unknown scenario are out of the question.

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Resources

Web resources

You can also visit the **Studio Web - Official documentation** (https://docs.uipath.com/studio-web/automation-cloud/latest/user-guide/overview) page to learn more about what you can do with Studio Web!

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